

Precedent

Every incident resolved becomes precedent.

Team — UK AI Agent Hackathon EP5 · Conduct “Make Legacy Move” · Fetch.ai · BasedAI

Something breaks → find the manual → click through an admin console.

8.85 hours

average time to resolve an incident — for a fix that is often minutes of keystrokes once it's known

\$600B / year

downtime cost across the Global 2000 — up 50% in two years (Splunk, 2026)

≈ **\$200M per company (Oxford Economics/Splunk, 2024)**

>60% of incidents are repeats — the fix already exists, nobody can find it (ServiceNow's own support org)

AI SREs fix broken code. In real enterprises, the fix is almost never code — it's a documented admin change.

Precedent remembers every fix your organisation has ever applied — and applies it again.

risk-classified · approval-gated · audited · reversible

detect → find the documented fix → check risk → get approval → execute → verify → remember

Same incident. Same fix.

Manual: ticket → search KB → admin console → approval queue → resolve

8.85 hrs

First time with Precedent (human approves) ~60 s



Every time after (pre-approved) ~15 s



The second time is free.

MediaCo — a simulated broadcaster · real data · live Jira Service Management · unscripted inputs

1. Broken EPG publish → plan + rollback → human approves → fixed in ~60 s, Jira ticket closes itself
2. **Same class recurs → pre-approved standard change → fixed in ~15 s**
3. Rights conflict, restricted runbook → Precedent refuses and routes to the right team

Seed data: real public runbooks (GitLab, Kubernetes SIG, the published CrowdStrike remediation), real programme metadata, and a 141k-event public IT-incident log ingested as fix history. Incident text is mutated at generation time — typos, vague symptoms, missing codes.

Autonomy is earned per incident-class — and a human grants it.

L0 Observe → L1 Recommend → L2 Execute after approval → **L3 Standing Approval**
(pre-approved standard change)

Promote to standing approval

Revoke

Approval moves earlier in time. It never leaves the loop.

Five specialised agents · deterministic policy engine · permission-aware memory



Live Jira Service Management · open-weight models end-to-end · agents live on Fetch.ai Agentverse / ASI:One

AI SREs stop at diagnosis. RPA is incident-blind. Runbook automation only runs pre-written scripts. ServiceNow deflects tickets inside its own walls.

The closed loop — retrieve your documented fix → approval-gated execution in third-party systems → verified, audited, remembered — is unclaimed.

ServiceNow paid \$2.85B for Moveworks — resolution memory is worth buying.

Every resolution writes a record: symptom → verified fix → approver → risk → rollback → outcome.

incident → fix → precedent → faster next time → more incidents trusted to it → ♡

141,000 real incident events ingested · a 25k-record incident store

94% arrived with their fix already precedent — and still took a median 18 hrs (calendar) to resolve by hand

Permission-checked retrieval: P99 0.445 μ s · FNR 0 leaks / 5,219 deny-expected · 6/6 adversarial attacks

Triage on mangled tickets (100-mutation corpus): 0 false-fast-paths / 100 · 25/25 red-herring decoys resisted (8% correct-match, 50% safe-degrade, 0% false fast-path)

Media ops first — regulated failure (Ofcom), 24/7 deadlines, legacy consoles

One MSP deal = hundreds of channels — Encompass runs 1,200+ channels/day; Amagi 5,000+ deliveries (vendor-claimed)

Then: every ops team drowning in runbooks — every ticket deflected saves \$22 → \$69 → \$104 per escalation tier

Agents made retrieve-and-safely-execute possible this year. Incumbents are priced to meter the workflow, not delete it.

Founding team — UK AI Agent Hackathon EP5

(one of us watched the incident loop first-hand inside broadcast-streaming operations)

The ask: two intros to broadcast-ops or MSP design partners — and we're applying to Antler & EWOR with this.

ServiceNow paid \$2.85B for resolution memory. We're building the version that executes.

A1 — Every number, sourced

Global 2000 downtime bill **\$600B/yr, +50% in 2 yrs** Splunk 2026 (exact framing)

Per-company downtime cost **\$200M/yr** Oxford Economics/Splunk 2024 — verified 3–0

Repeat incidents with existing fixes **>60%** ServiceNow KCS case study — verified 3–0

Cost per ticket L1 → desktop → L3 **\$22 → \$69 → \$104** MetricNet whitepaper (~2019–20)

Average incident MTTR **8.85 business hrs** MetricNet/HDI

KB-attached / KCS relief **66% faster / 52% faster** ServiceNow — verified

Moveworks acquisition **\$2.85B** ServiceNow newsroom, Mar 2025

Deflection precedent **50–88%** Moveworks — vendor-claimed

Fix-class match rate (24,918 real incidents) **94.4% (symptom 98.6%)** our measurement, UCI CC BY 4.0

Median resolution, precedented repeats **18.2 calendar hrs (p75 136.6)** ours — never blend w/ 8.85 business

Recurring classes ≥4 occurrences **558 classes = 94.8% of volume** ours — ladder-bootstrap evidence

Extractor robustness (100-mutation corpus) **0 false-fast-paths / 100; 25/25 decoys resisted** ours — seed 4207

Conformance FNR / attacks **0 leaks / 5,219; 6/6** ours — independent-oracle bench

A2 — Data provenance (the “not made up” slide)

KB seeded with real published runbooks (GitLab, Kubernetes SIG, the CrowdStrike channel-file remediation) — every article carries its adapted_from URL.

Fix history from the UCI ServiceNow incident event log (141,712 events / 24,918 incidents, CC BY 4.0) — measured 94% fix-class match, 18.2h calendar median for precededented repeats.

Scheduler/rights/EPG seeded with real UK programme metadata (TVmaze CC BY-SA / Freeview XMLTV) and real streaming-catalog data (CC0 Kaggle) — only licence-window terms are synthesised, by a stated rule.

We checked licences and REJECTED TMDb (API terms prohibit AI/ML use) and IMDb — the diligence is itself a credibility beat.

Incident text mutated at generation time (typos, colloquial symptoms, missing codes, red herrings). The systems are simulated; the content is real.

A3 — Standing approval: exact semantics

Incident-class key = deterministic fingerprint sha256(service | error_code | target_object_type) from structured fields — the LLM proposes candidates, but a match only counts on fingerprint equality. The model never authorises itself.

Graduation, printed in the audit log: 3 consecutive verified L2 successes, zero rollbacks → eligible; a human clicks Promote to Standing Approval. Any verification failure or rollback auto-demotes the class to L1 and logs a demotion event.

The rollback plan is generated BEFORE execution and is a precondition of the gate; verification failure fires it automatically.

Approver identity is recorded per action (chat: the Chat Protocol sender address; production: SSO identity — the control structure is identical).

A4 — Permission-aware memory (BasedAI depth)

Every record stores the FULL set of source permission constraints; retrieval must satisfy ALL of them — conjunction, not one strictest label.

A precompiled effective-policy cache makes the check one indexed lookup: P99 0.445 μ s (end-to-end overhead P99 0.0130 ms) over the 25k-record store, concurrent queries.

Benchmarked in the track's own vocabulary: FNR 0 / 5,219 and FPR 0 / 4,781 over 10,000 ground-truth queries graded by an INDEPENDENT oracle, P50/P99, overhead, ACL drift, time-to-consistency — each vs threshold in bench/RESULTS.md; six named attacks pass.

Revocation cascades — dual enforcement: flip the issue security level in Jira → Jira hides the runbook AND every derived fix/summary/embedding becomes unretrievable within one poll tick, with denial audit events in BOTH logs.

Fallback mode fails closed: uncertain ACL freshness → not served. Ships as a standalone library; 100% open-weight models, named in the README.

A5 — Why won't ServiceNow / Conduct build this?

ServiceNow monetises seats + tickets; auto-resolution cannibalises both. They ACQUIRE this layer: Jeli \$29.7M, Moveworks \$2.85B. We're building what they buy.

Now Assist acts inside ServiceNow workflows; we execute inside third-party business-app admin surfaces — their walls are our territory.

Conduct makes legacy legible; we make it operable — a complement, and the brief told us to build exactly this (“start with the process”).

Rehearsed: “They meter the workflow; we delete it. Incumbents don't ship products that shrink their own ticket count — they buy them. That's the Moveworks story.”

A6 — Integration reality (the “you built the simulator” defence)

Four-tier surface strategy: REST APIs (WhatsOn-class schedulers) → BXF/file exchange → watched-folder/FTP drops → RPA-style console driving (last resort, same approval gates).

The Operator executes only TYPED tool calls per surface — never free-form shell; tier 3/4 fixes start at L0/L1 and stay approval-gated.

MediaCo's endpoints deliberately mirror the surfaces the real vendors document (Grass Valley STRATUS: “traffic integration via BXF files”).

A7 — Fetch.ai deliverables

Three agents registered on Agentverse as mailbox agents, collaborating over the uAgents Agent Chat Protocol; both Innovation Lab + hackathon badges on each profile; the hosted degraded-L0 Watcher stays live post-hackathon.

The full loop runs inside an ASI:One conversation: report an incident in chat → plan + rollback → “approve” → the ticket transitions and closes → audit hash returned.

Approval principal = the Chat Protocol sender address, logged as the authorising identity, with a 10-minute gate TTL (a dropped session never leaks an execution).

Registered · active on Agentverse (Almanac-confirmed): Watcher agent1q2m0gk9...jcgkldx · Librarian agent1qv760pr2...kse9xv7 · Operator agent1qwesj8x7...uxna02g.

Full profile URLs + the ASI:One shared-chat URL are listed in the submission (Fetch deliverables).

A9 — Bottoms-up ACV

One arithmetic slide, every input sourced:

1 MSP NOC (Encompass-class, 1,200+ channels) × tens of thousands of tickets/yr × 60%+ repeat share (ServiceNow, verified 3–0; our corpus says 94% precededent) × \$50 blended deflection value (inside the \$22–\$104 MetricNet ladder)

⇒ **seven-figure ACV justification per site — before counting downtime avoidance.**

This is the bottoms-up serviceable-market answer; it does not replace primary customer evidence and never pretends to.

A8 — Liability & regulated ops

Precedent executes only documented, previously-verified fixes at the customer's chosen autonomy level — the customer's own change-management policy, encoded and enforced. Shadow-mode-first at MSPs.

Headline metric we quote: per-class execution success + rollback rate from real runs — never model benchmarks (the incident.io-vs-Rootly accuracy-theatre war is the cautionary tale).

Every action carries: input, reasoning, confidence, action, authorising rule + approver — the audit schema regulators and SOC 2 auditors already recognise.